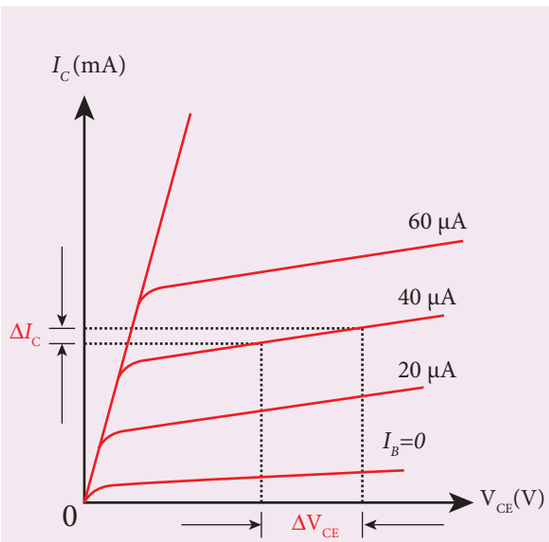
### (ii) Output characteristic curve: $V_{CE}$ vs $I_C$ ($I_B$ constant)

- The base current $I_B$ is kept constant.
- $V_{CE}$ is varied in steps of 1V and the corresponding collector current $I_C$ is noted. The readings are taken till the collector current becomes almost constant.
- Initially $I_B$ is kept at 0 mA and the corresponding collector current is noted. This current is the reverse saturation current $I_{CEO}$.
- The experiment is repeated for various values of $I_B$. The readings are tabulated.
- A graph is drawn by taking $V_{CE}$ along x-axis and $I_C$ along y-axis for various values of $I_B$.
- The set of curves thus obtained is called the output characteristics of a transistor.
- The reciprocal of the slope of the curve gives output impedance of the transistor.

| S. No | $I_B$ =20 μA | | $I_B$ =40 μA | |
|---|---|---|---|---|
| | $V_{CE}$ (V) | $I_C$ (mA) | $V_{CE}$ (V) | $I_C$ (mA) |
| | | | | |




$$\text{Slope} = \frac{\Delta I_C}{\Delta V_{CE}}$$

$$\text{Output impedance} = \frac{1}{\text{slope}}$$

### (iii) Transfer characteristic curve: $I_B$ vs $I_C$ ($V_{CE}$ constant)

- The collector-emitter voltage $V_{CE}$ is kept constant.
- The base current $I_B$ is varied in steps of 10 μA and the corresponding collector current $I_C$ is noted.
- This is repeated by changing the value of $V_{CE}$. The readings are tabulated.
- The transfer characteristics is a plot between the input current IB along x-axis and the output current $I_C$ along y-axis keeping $V_{CE}$ constant.
- The slope of the transfer characteristics plot gives the current gain β can be calculated.



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